As one reward for his contributions to Soviet nuclear might (Chapter 6), Zel'dovich had been given one of 2B's eight apartments: the southwest quarter of the second floor. The apartment was enormous by Moscow standards, 1500 square feet; he shared it with his wife, Varvara Pavlova, one daughter, and a son-in-law.

Zel'dovich met me at the apartment door, with a warm grin on his face and the sounds of his bustling family emerging from back rooms. I removed my shoes, put on slippers from the pile beside the door, and followed him into the shabby but comfortable living/dining room, with its overstuffed couch and chairs. On one wall was a map of the world, with colored pins identifying all the places to which Zel'dovich had been invited (London, Princeton, Beijing, Bombay, Tokyo, and many more), and which the Soviet state, in its paranoid fear of losing nuclear secrets, had forbade him to visit.

Zel'dovich, his eyes dancing, sat me down at the long dining table dominating the room's center, and announced, "A spinning black hole must radiate. The departing radiation will kick back at the hole and gradually slow its spin, and then halt it. With the spin gone, the radiation will stop, and the hole will live forever thereafter in a perfectly spherical, nonspinning state."

"That's one of the craziest things I've ever heard," I asserted. (Open confrontation is not my style, but Zel'dovich thrived on it. He wanted it, he expected it, and he had brought me to Moscow in part to serve as a sparring partner, an opponent against whom to test ideas.) "How can you make such a crazy claim?" I asked. "Everyone knows that radiation can flow into a hole, but nothing, not even radiation, can come out."

Zel'dovich explained his reasoning: "A spinning metal sphere emits electromagnetic radiation, and so, similarly, a spinning black hole should emit gravitational waves."

A typical Zel'dovich proof, I thought to myself. Pure physical intuition, based on nothing more than analogy. Zel'dovich doesn't understand general relativity well enough to compute what a black hole should do, so instead he computes the behavior of a spinning metal sphere, he then asserts that a black hole will behave analogously, and he wakes me up at 6:30 A.M. to test his assertion.

However, I had already seen Zel'dovich make discoveries with little more basis than this; for example, his 1965 claim that when a mountainous star implodes, it produces a perfectly spherical black hole (Chapter 7), a